

achieves hardware-aligned sparse attention computation throughout the full model lifecycle, including training, prefilling, and decoding.

8. Conclusion

We present NSA, a hardware-aligned sparse attention architecture for efficient long-context modeling. By integrating hierarchical token compression with blockwise token selection within a trainable architecture, our architecture achieves accelerated training and inference while maintaining Full Attention performance. NSA advances the state-of-the-art by demonstrating general benchmark performance matches full-attention baselines, exceeding modeling capability in long-context evaluations, and enhanced reasoning ability, all accompanied by measurable reductions in computational latency and achieving significant speedup.

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